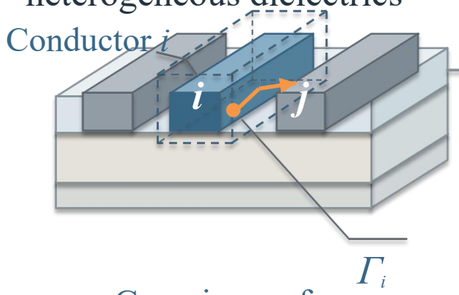
(a) Physical input

Input

3-D conductors and heterogeneous dielectrics



Master conductor i

Representative local dielectric states



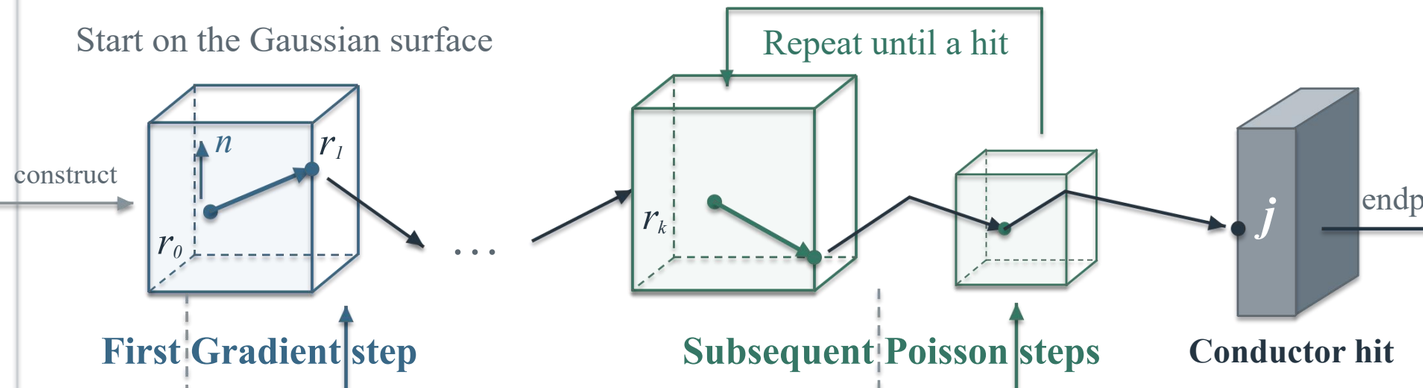
State-dependent transition behavior

$$X(r) \in (0, 1]^{23\times23\times23}$$

Normalized dielectric tensor at the current walk state

Local state

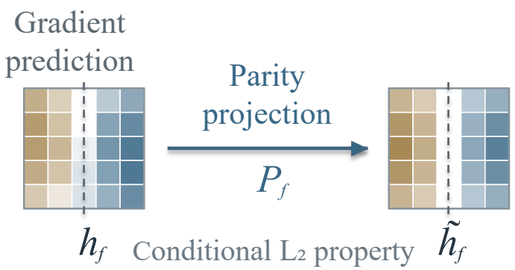
(b) CARE-RWCap refinement within neural-guided FRW



1 Conditional Parity Gradient Refinement Component

Reflection & directional structure

Strict check · face-pair averaging



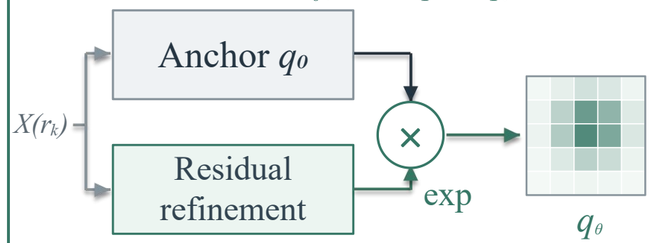
$$\beta_f = \|\tilde{h}_f\|_1 / \|h_f\|_1$$

L1-ratio contribution compensation

2 Condition-Aware Poisson Refinement Component

In-face representation

Retain π_θ^P; refine $q_0 \rightarrow q_\theta$



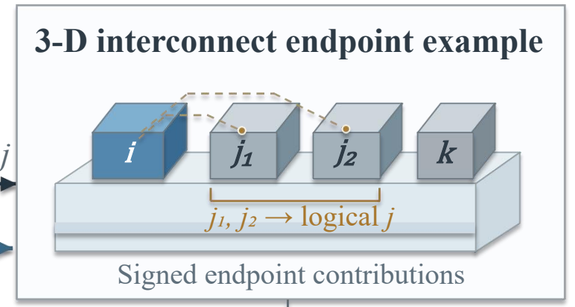
$\tilde{r}_\theta$, $0 < a_\theta < \tau$ Reweight + normalize

$$q_\theta \propto q_0^{\varepsilon} \exp(a_\theta \tilde{r}_\theta)$$

Offline loss: KL · action · trust · tail risk

Signed path weight $\omega^{(m)}$

(c) Same-solve readout



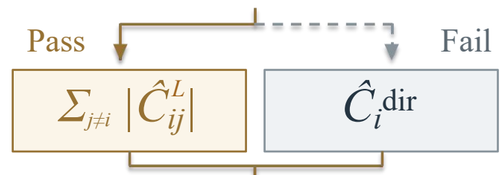
Logical-conductor row

$\hat{C}_{ii}^L$ $\hat{C}_{ij}^L$ $\hat{C}_{ik}^L$ …

3 Conditional Endpoint Re-estimation Component

Conditional self-capacitance readout

Complete, finite logical row?
Nonpositive couplings; positive coupling sum?



Output: capacitance matrix $\hat{C}$

